2021 Summer Research Diary

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Summer 2021

Week 1: May 5

- Meeting the team
- Project introduction
- Software installation
- Literature Review (Ongoing)

Setting Up IKEA Furniture Assembly Environment [1]

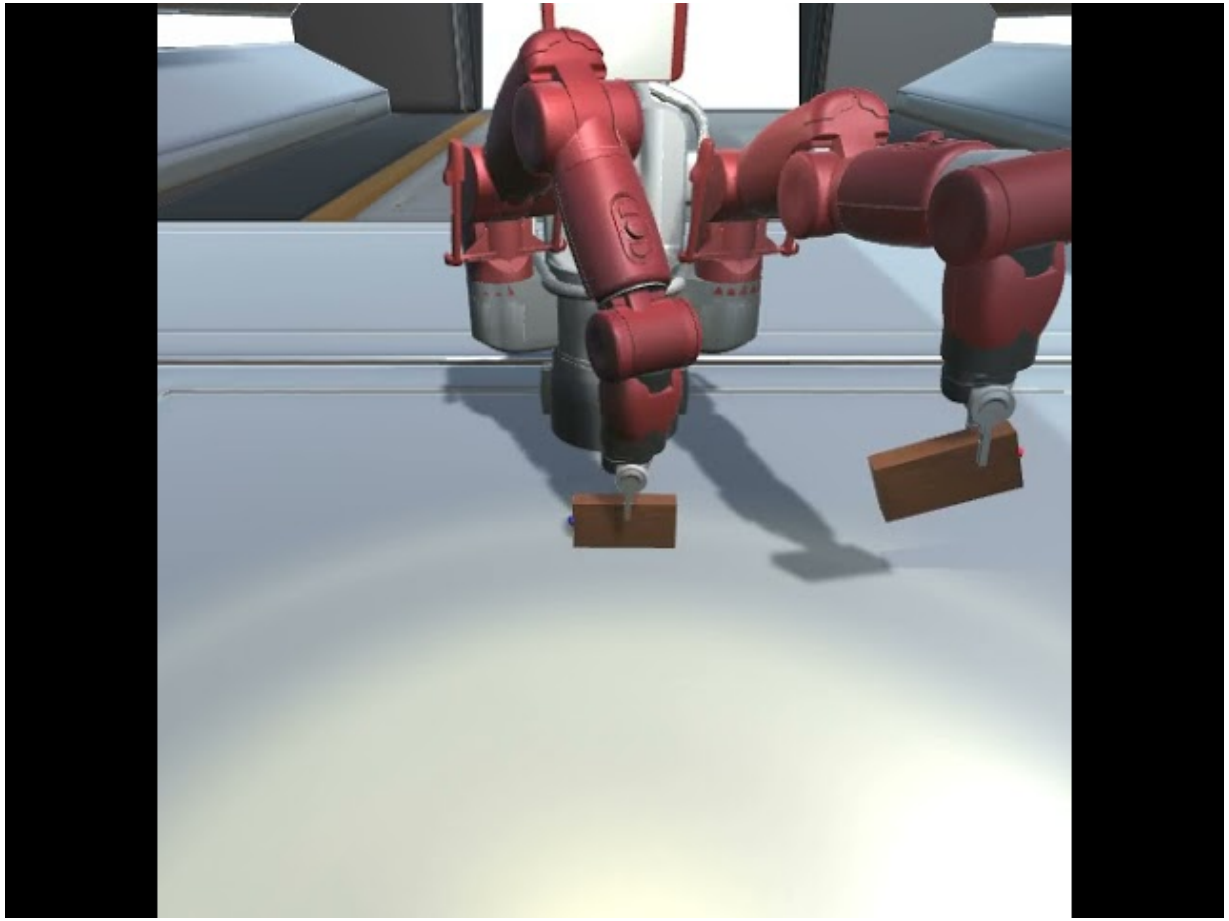

Figure: Manual control of BAXTER arms using the IKEA furniture assembly environment from [1]

Task and Motion Planning Literature Review

- IKEA Furniture Assembly Environment for Long-Horizon Complex Manipulation Tasks [1]
- Integrated Task And Motion Planning [2]
- Learning Heuristic Functions from Relaxed Plans [3]
- Learning Domain-Independent Planning Heuristics with Hypergraph Networks [4]

Week 2: May 12

- Catchup tutorial with Mohamed (software setup and debug)
- Integrating Task and Motion Planning Tutorial [5]
- Looking for the appropriate inverse kinematics solver
- **Inverse Kinematics and Motion Planning In Drake**

Inverse Kinematics

Constraints on IK algorithm:

- Can handle ≥ 7 degrees of freedom (without just sampling for the 7th).
- Respects joint limits.
- Generic task space constraints (avoiding collisions with objects on table) [6].
- Faster is better.

Solution: **Drake**

IK with Drake

- Mathematical programming means we can handle arbitrary:
  - DOF.
  - Constraints.
  - Cost functions.
- Automatically selects appropriate solver
- Clean and fast simulation environment:
  - Supporting URDF/SDF.
  - Already includes model for Franka Panda arm.

Formulating the IK problem

- Given:
  - A trajectory (list of poses) of the end-effector.
  - A list of objects to avoid collision with.
  - Nominal (centered) joint positions (that are preferable).
- Constraints, costs, and initial guess:
  - End-effector position is within a bounding box of the desired position.
  - End-effector orientation is within some angle of the desired orientation.
  - Quadratic-error cost for (joint) distance from nominal positions.
  - Set initial guess for current solution to the solution previous solution (or the nominal joint positions). This makes the solutions connect.
  - Add minimum distance constraint between the arm links and the objects we don't want to collide with.

Simulating IK in Drake

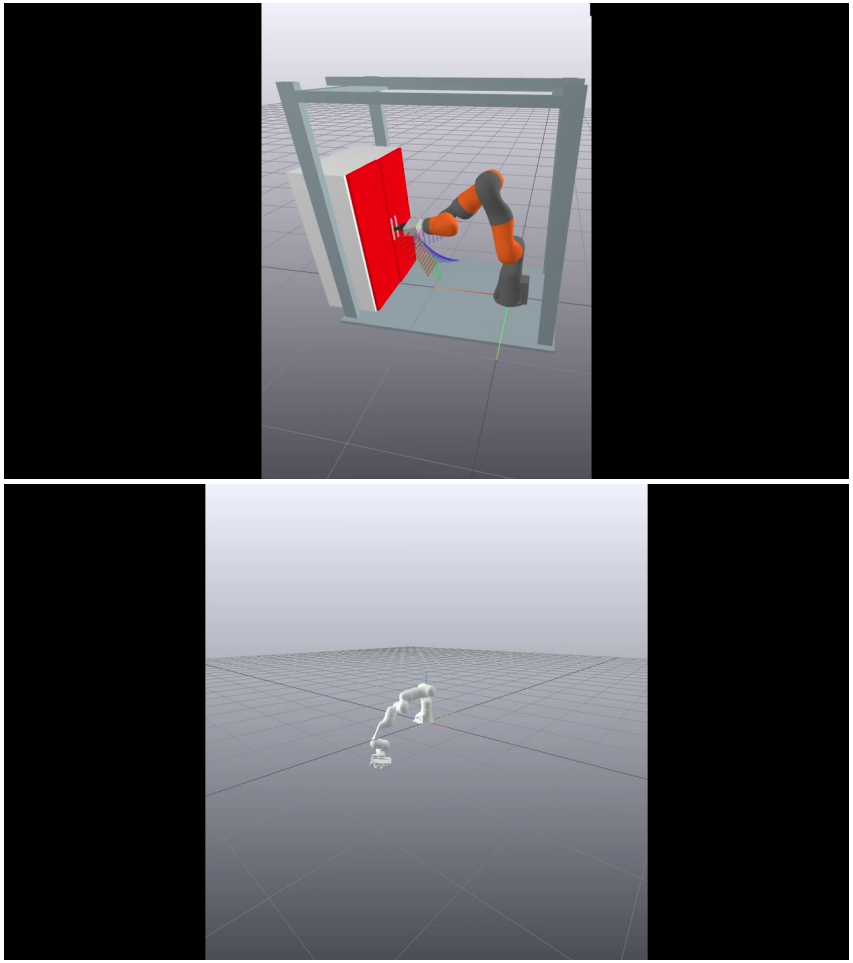

Figure: IK for a given trajectory (top) and simulation of passive dynamics (bottom) [7] [8].

Google collab with IK widget:
click here

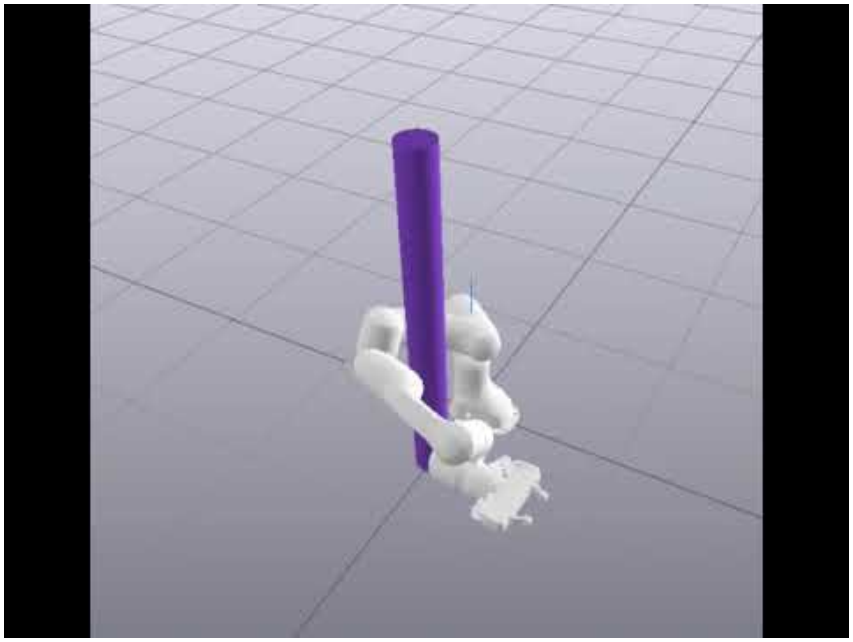

Figure: Teleoperated IK with collision avoidance.

Next Steps

- Plugging drake into existing furniture simulation.
  - Or porting furniture assembly simulation to Drake (maybe the better option in the long-term)?
- More work to control the robot arm properly in simulation:
  - Sending signals to joint actuators.
  - Controlling finger joints to pick things up.
- More exploration of methods to find best kinematics solutions (e.g RRT).

References I

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References III

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